

PL SQL EXERCISES

1. Control Structure

```
SET SERVEROUTPUT ON;

DECLARE
num NUMBER := 7;
grade CHAR := 'B';
i NUMBER := 1;
BEGIN
DBMS_OUTPUT.PUT_LINE('--- IF Statement ---');

IF num > 0 THEN
DBMS_OUTPUT.PUT_LINE(num || ' is Positive');
END IF;

DBMS_OUTPUT.PUT_LINE(CHR(10) || '--- IF-ELSIF-ELSE Statement ---');

IF num < 0 THEN
DBMS_OUTPUT.PUT_LINE('Negative Number');
ELSIF num = 0 THEN
DBMS_OUTPUT.PUT_LINE('Zero');
ELSE
DBMS_OUTPUT.PUT_LINE('Positive Number');
END IF;

DBMS_OUTPUT.PUT_LINE(CHR(10) || '--- CASE Statement ---');

CASE grade
WHEN 'A' THEN
DBMS_OUTPUT.PUT_LINE('Excellent');
WHEN 'B' THEN
DBMS_OUTPUT.PUT_LINE('Very Good');
WHEN 'C' THEN
DBMS_OUTPUT.PUT_LINE('Good');
ELSE
DBMS_OUTPUT.PUT_LINE('Needs Improvement');
END CASE;

DBMS_OUTPUT.PUT_LINE(CHR(10) || '--- FOR LOOP ---');

FOR j IN 1..5 LOOP
DBMS_OUTPUT.PUT_LINE('FOR LOOP: ' || j);
END LOOP;

DBMS_OUTPUT.PUT_LINE(CHR(10) || '--- WHILE LOOP ---');

WHILE i <= 5 LOOP
DBMS_OUTPUT.PUT_LINE('WHILE LOOP: ' || i);
i := i + 1;
END LOOP;

DBMS_OUTPUT.PUT_LINE(CHR(10) || '--- SIMPLE LOOP with EXIT ---');

i := 1;

LOOP
DBMS_OUTPUT.PUT_LINE('LOOP: ' || i);
i := i + 1;

EXIT WHEN i > 5;
```

```

END LOOP;

DBMS_OUTPUT.PUT_LINE(CHR(10) || '--- CONTINUE Statement ---');

FOR k IN 1..5 LOOP
CONTINUE WHEN k = 3;
DBMS_OUTPUT.PUT_LINE('CONTINUE LOOP: ' || k);
END LOOP;

END;
/

```

OUTPUT

```

--- IF Statement ---
7 is Positive
--- IF-ELIF-ELSE Statement ---
Positive Number
--- CASE Statement ---
Very Good
--- FOR LOOP ---
FOR LOOP: 1
FOR LOOP: 2
FOR LOOP: 3
FOR LOOP: 4
FOR LOOP: 5
--- WHILE LOOP ---
WHILE LOOP: 1
WHILE LOOP: 2
WHILE LOOP: 3
WHILE LOOP: 4
WHILE LOOP: 5

```

```

--- SIMPLE LOOP with EXIT ---
LOOP: 1
LOOP: 2
LOOP: 3
LOOP: 4
LOOP: 5

--- CONTINUE Statement ---
CONTINUE LOOP: 1
CONTINUE LOOP: 2
CONTINUE LOOP: 4
CONTINUE LOOP: 5

PL/SQL procedure successfully completed.

```

2. Stored Procedures

-- Enable Output

```
SET SERVEROUTPUT ON;
```

```

-----
-- STEP 1 : CREATE TABLE
-----

```

```
CREATE TABLE EMPLOYEE_DEMO
(
    EMP_ID NUMBER PRIMARY KEY,
    EMP_NAME VARCHAR2(50),
    SALARY NUMBER
);
```

```
-----
-- STEP 2 : INSERT SAMPLE DATA
-----
```

```
INSERT INTO EMPLOYEE_DEMO VALUES(101,'Pavithra',35000);
INSERT INTO EMPLOYEE_DEMO VALUES(102,'Raghavi',55000);
INSERT INTO EMPLOYEE_DEMO VALUES(103,'Rakshita',75000);
```

```
COMMIT;
```

```
-----
-- STEP 3 : PROCEDURE WITHOUT PARAMETERS
-----
```

```
CREATE OR REPLACE PROCEDURE WELCOME_MESSAGE
IS
BEGIN
    DBMS_OUTPUT.PUT_LINE('-----');
    DBMS_OUTPUT.PUT_LINE('WELCOME TO PL/SQL STORED PROCEDURES');
    DBMS_OUTPUT.PUT_LINE('-----');
END;
/
```

```
-----
-- STEP 4 : PROCEDURE WITH IN PARAMETER
-----
```

```
CREATE OR REPLACE PROCEDURE DISPLAY_EMPLOYEE
(
```

```

P_EMPID IN NUMBER
)
IS
V_NAME EMPLOYEE_DEMO.EMP_NAME%TYPE;
V_SALARY EMPLOYEE_DEMO.SALARY%TYPE;
BEGIN

SELECT EMP_NAME,SALARY
INTO V_NAME,V_SALARY
FROM EMPLOYEE_DEMO
WHERE EMP_ID=P_EMPID;

DBMS_OUTPUT.PUT_LINE('Employee ID : '||P_EMPID);
DBMS_OUTPUT.PUT_LINE('Employee Name : '||V_NAME);
DBMS_OUTPUT.PUT_LINE('Salary      : '||V_SALARY);

END;
/

```

```

-----
-- STEP 5 : PROCEDURE WITH OUT PARAMETER
-----

```

```

CREATE OR REPLACE PROCEDURE GET_SALARY
(
P_EMPID IN NUMBER,
P_SALARY OUT NUMBER
)
IS
BEGIN

SELECT SALARY
INTO P_SALARY
FROM EMPLOYEE_DEMO
WHERE EMP_ID=P_EMPID;

```

END;

/

-- STEP 6 : PROCEDURE WITH IN OUT PARAMETER

CREATE OR REPLACE PROCEDURE INCREMENT_SALARY

(

 P_SALARY IN OUT NUMBER

)

IS

BEGIN

 P_SALARY:=P_SALARY+5000;

END;

/

-- STEP 7 : PROCEDURE USING IF ELSE

CREATE OR REPLACE PROCEDURE SALARY_GRADE

(

 P_SALARY IN NUMBER

)

IS

BEGIN

 IF P_SALARY>=70000 THEN

 DBMS_OUTPUT.PUT_LINE('Grade : A');

 ELSIF P_SALARY>=50000 THEN

```

        DBMS_OUTPUT.PUT_LINE('Grade : B');

ELSE

        DBMS_OUTPUT.PUT_LINE('Grade : C');

END IF;

END;

/

-----
-- STEP 8 : EXECUTE ALL PROCEDURES
-----

DECLARE

V_SALARY NUMBER;
V_INCREMENT NUMBER:=30000;

BEGIN

        DBMS_OUTPUT.PUT_LINE(' ');
        DBMS_OUTPUT.PUT_LINE('***** PROCEDURE WITHOUT PARAMETER *****');
        WELCOME_MESSAGE;

        DBMS_OUTPUT.PUT_LINE(' ');
        DBMS_OUTPUT.PUT_LINE('***** PROCEDURE WITH IN PARAMETER *****');
        DISPLAY_EMPLOYEE(102);

        DBMS_OUTPUT.PUT_LINE(' ');
        DBMS_OUTPUT.PUT_LINE('***** PROCEDURE WITH OUT PARAMETER *****');
        GET_SALARY(101,V_SALARY);
        DBMS_OUTPUT.PUT_LINE('Salary Returned : '||V_SALARY);

        DBMS_OUTPUT.PUT_LINE(' ');

```

```

DBMS_OUTPUT.PUT_LINE('***** PROCEDURE WITH IN OUT PARAMETER *****');
DBMS_OUTPUT.PUT_LINE('Salary Before Increment : '||V_INCREMENT);
INCREMENT_SALARY(V_INCREMENT);
DBMS_OUTPUT.PUT_LINE('Salary After Increment : '||V_INCREMENT);

DBMS_OUTPUT.PUT_LINE(' ');
DBMS_OUTPUT.PUT_LINE('***** PROCEDURE USING IF ELSE *****');
SALARY_GRADE(V_SALARY);

END;
/

```

OUTPUT

```

***** PROCEDURE WITHOUT PARAMETER *****

-----
WELCOME TO PL/SQL STORED PROCEDURES
-----

***** PROCEDURE WITH IN PARAMETER *****

Employee ID   : 102
Employee Name : Raghavi
Salary        : 55000

***** PROCEDURE WITH OUT PARAMETER *****

Salary Returned : 35000

***** PROCEDURE WITH IN OUT PARAMETER *****

Salary Before Increment : 30000
Salary After Increment  : 35000

***** PROCEDURE USING IF ELSE *****

Grade : C

PL/SQL procedure successfully completed.

```